Graph Theory

What is a graph?

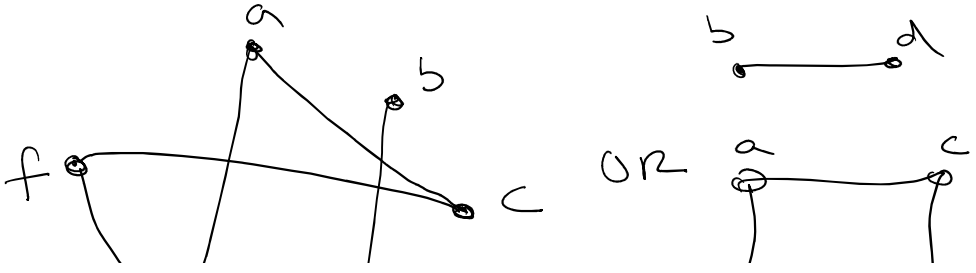
A graph G is a tuple, i.e., $G := (V, E)$, where V (or $V(G)$) is a set of elements called the vertex set (or node set) (each element in this set is called a vertex (or node)) and E (or $E(G)$) is a collection of 2-sized subsets of V and is called the edge set (each element in E is called an edge).

Example

$G = (V, E)$, where

$$V = \{a, b, c, d, e, f\}$$

$$E = \{\{a, c\}, \{a, e\}, \{b, d\}, \{e, f\}, \{c, f\}\}$$





A more general notion

Hypergraph → is a tuple

$H = (V, E)$

V → vertex set (a collection of elements)

E → a collection of subsets of V.

(a subset in E is called a hyperedge)

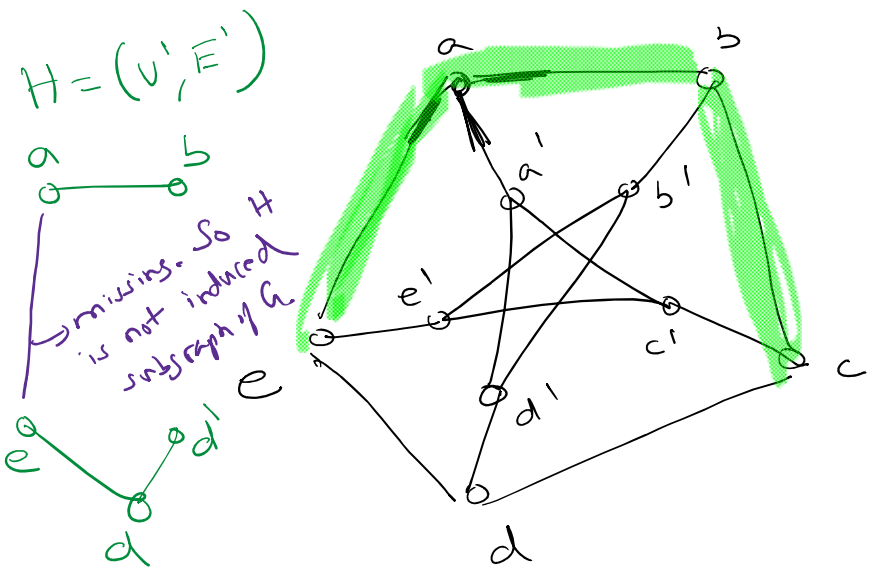
Example

$H = (V, E)$

$V = \{a, b, c, d, e, f\}$

$E = \{\{a, b, c, d\}, \{c, e, f\}, \{a, e, f\}, \{b, d\}\}$

Some definitions on graphs



$G = (V, E)$

$V = \{a, b, c, d, e, a', b', c', d', e'\}$

$E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{d, e\}, \{e, a\}, \{a, a'\}, \{b, b'\}, \ldots\}$

OR

(a, b)



$|V| = 10$

$|E| = 15$

a is a neighbour of b

$N_G(a) = \{b, a', e\}$

OR (a,b) $\{a,a'\}$, $\{b,b'\}$, ...

OR ab $\{e,e'\}$, $\{a'',c'\}$, $\{c',e''\}$, $\{e',b'\}$, $\{b',d'\}$, $\{d',a'\}$ }

Notation

(i) degree of a vertex $\longrightarrow$ no. of edges it is present in.

$$\deg_G(a) = 3, \quad \deg_G(b) = 3$$

(ii) maximum degree of G, denoted by $\Delta(G)$, is

$$\Delta(G) := \max\{\deg_G(v) : v \in V(G)\}$$

above example, $\Delta(G) = 3$.

(iii) minimum degree of G, denoted by

$$\delta(G) := \min\{\deg_G(v) : v \in V(G)\}$$

above example, $\delta(G) = 3$

(iv) given two vertices u and v, we say u is adjacent to v

if $\{u,v\} \in E(G)$.

u is a neighbor of v in G if $\{u,v\} \in E(G)$.

(v) Neighborhood of a vertex v in G is denoted $N_G(v)$ and is defined as $N_G(v) = \{u : \{u,v\} \in E(G)\}$

(vi) $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

(vii) a vertex v is incident with an edge e if $v \in e$.
↳ 2-sized subset of V.

(ix) $H = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' = \{(u,v) \in E(G) : u, v \in V'\}$

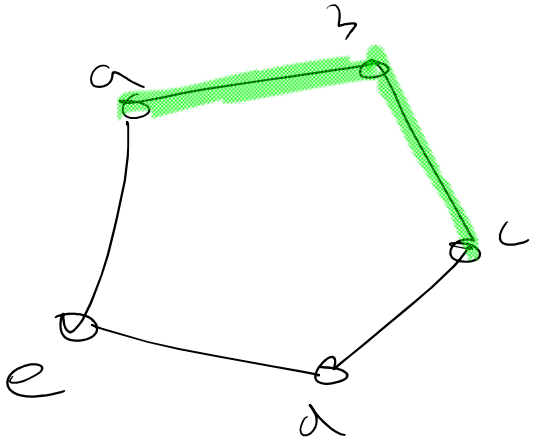
(x) $H = (V', E')$ is a spanning subgraph

of $G = (V, E)$ if H is a subgraph of G and $V' = V$.

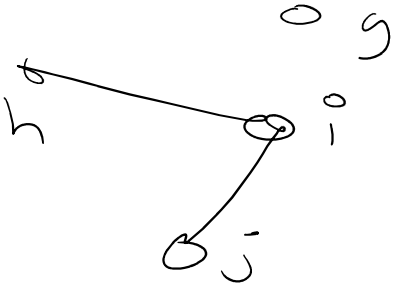
↳ spanning

(xi) Average degree of a graph G, denoted $d_{avg}(G)$ is

$$d_{avg}(G) := \frac{\sum_{v \in V(G)} deg_G(v)}{|V|}$$



for $H = (V, E)$



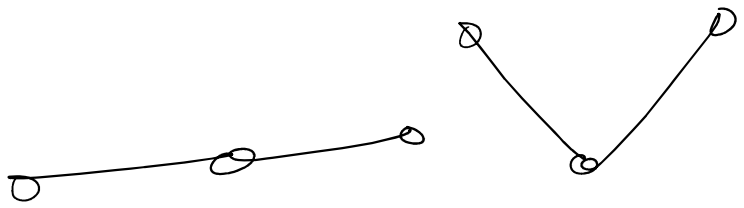


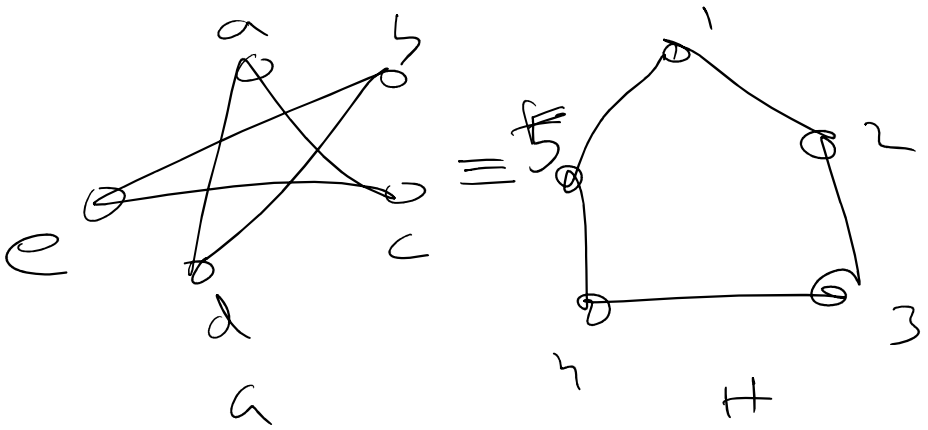
$V(H) = \{a, b, \ldots, l\}$

$E(H) = \{\{a, e\}, \{a, b\}, \ldots, \{k, l\}\}$

* sum of degrees $= 2 \cdot |E(G)|$.

Therefore no. of odd degree vertices is even.





G is isomorphic to H.

$f : V(G) \to V(G)$ as

$f(a) = 1$, $f(c) = 2$, $f(e) = 3$, $f(b) = 4$,

$f(d) = 5$

* A <u>path</u> is a non-empty graph

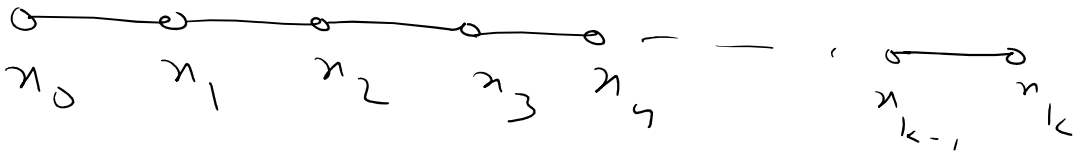
$P = (V, E)$, where

$V = \{x_0, x_1, x_2, \ldots, x_k\}$

$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \{x_2, x_3\}, \ldots$

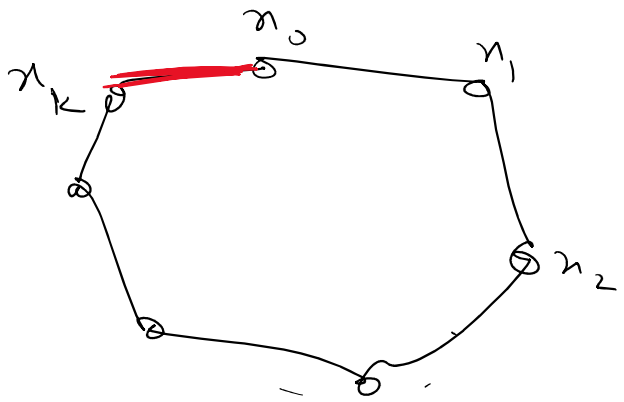
$\ldots \{x_{k-1}, x_k\}\}$

$\dots \{x_{k-1}, x_k\} \}$



length of a path is the no. of edges in it.

* (ycle



Let $P = (V, E)$ be a path,

where $V = \{x_0, x_1, \dots, x_k\}$

$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \dots, \{x_{k-1}, x_k\}\}$

Then, $P + \{x_k, x_0\}$ is a cycle

Lemma Every graph has (starting at a vertex with min degree)

(i) a path of length $\delta(G)$, and

(ii) a cycle of length $\delta(G) + 1$. $(\delta(G) \geq 2)$

(ii) a cycle of length $\delta(G)+1$. $(\delta(G) \geq 2)$

Proof.

Proving (i) Consider a graph G with min degree $\delta(G)$.

By mathematical induction on no. of vertices.

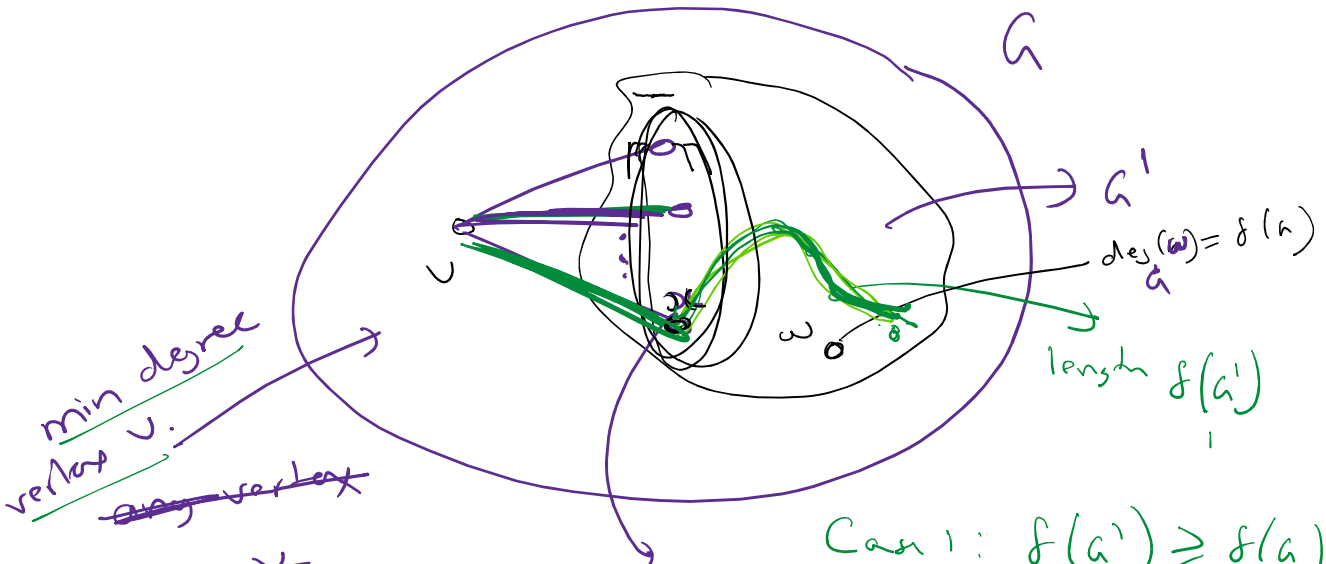
Base case: $|V(G)| = |V| = 1$.

$\delta(G) = 0$.

0-length path

Induction step. Assume the statement is TRUE for all graphs on at most n vertices.

Suppose G has $n+1$ vertices.



$\deg(v) = \delta(G)$.

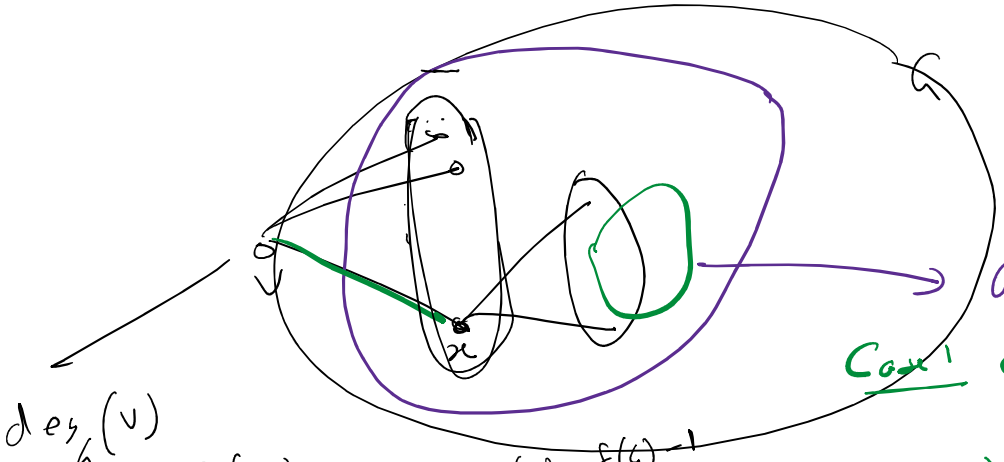
$\deg_G(x) = d(G)$

Case 1: $\delta(G') \geq \delta(G)$ good.

Case 2 $\delta(G') < \delta(G)$

$\deg_{G'}(x) = \delta(G) - 1$

Then, $\delta(G') = \delta(G) - 1$



$deg_G(v) = \delta(G)$ $\quad deg_{G'}(x) = \delta(G) - 1$

Case 1 $\delta(G') \geq \delta(G)$.

Good

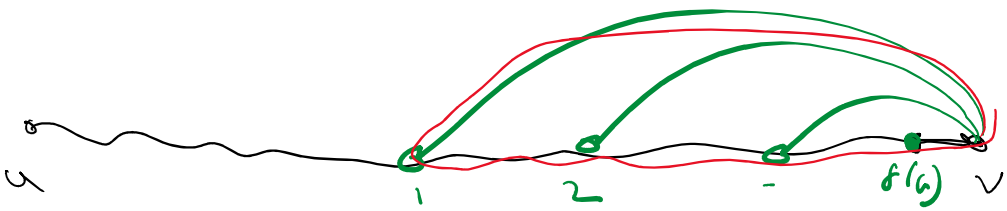
Case 2 $\delta(G') = \delta(G) - 1$

↳ How to go forward ??

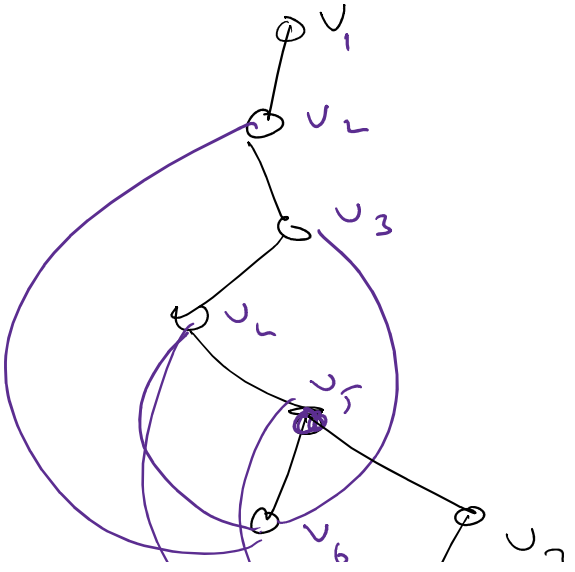
Alternate approach

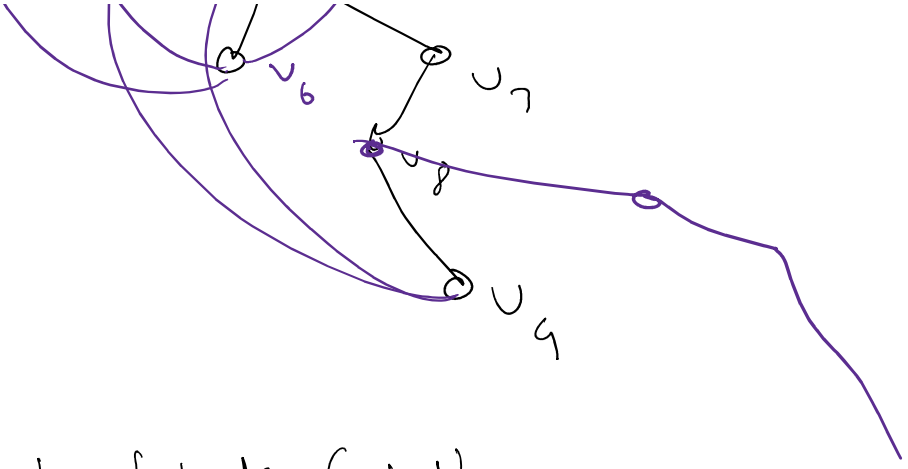
Proving (i) and (ii)

Take the longest path in G.



$deg_G(v) \geq \delta(G)$





Lecture by Subrata (Wed)

Thursday

Recap:

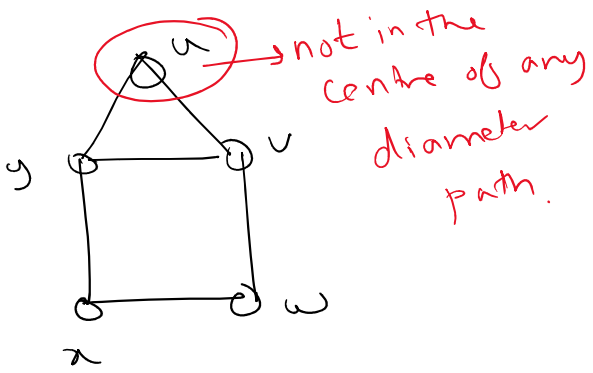
radius, diameter, central vertex, girth, circumferance, distance.

Stmt. Given a graph G, of G.

(i) Every central vertex is in the centre of a diameter path

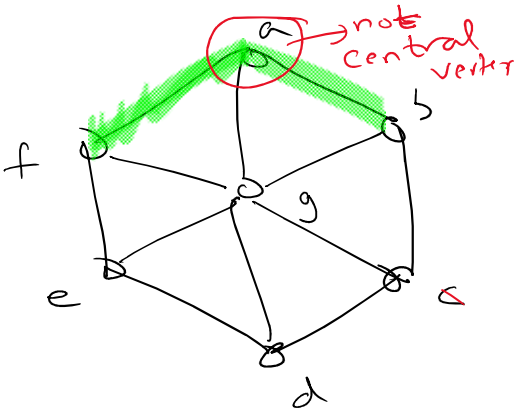
(ii) The centre of a diameter path is a central vertex of G.

Statement (i) is wrong



radius = 2

Statement (ii) is wrong



radius = 1

Diameter = 2
every vertex is a central vertex.

diameter = 2
g is a central vertex of G.

Proposition.

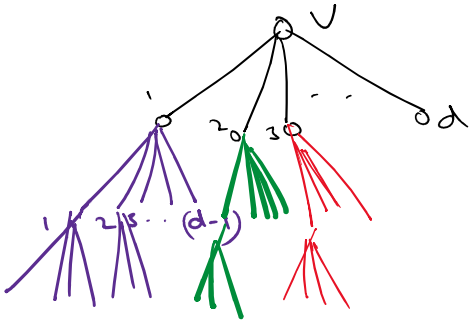
$$rad(G) \leq diam(G) \leq 2 \cdot rad(G).$$

Proposition $\quad radius(G) \leq k$

max deg of $G \leq d$, where $d \geq 3$

Show that the graph G has fewer than $\frac{d}{(d-2)}(d-1)^k$ vertices.

Proof: Let v be a central vertex of G.



Level 0 $\rightarrow$ 1

Level 1 $\rightarrow$ d

Level 2 $\rightarrow$ $d(d-1)$

Level 3 $\rightarrow$ $d(d-1)^2$

Level k $\rightarrow$ $d(d-1)^{k-1}$

$$\# \text{ vertices} \leq 1 + d + d(d-1) + \cdots + d(d-1)^{k-1}$$

$$\leq 1 + d\left(1 + (d-1) + \cdots + (d-1)^{k-1}\right)$$

$$= 1 + d\left(\frac{(d-1)^k - 1}{d-2}\right)$$

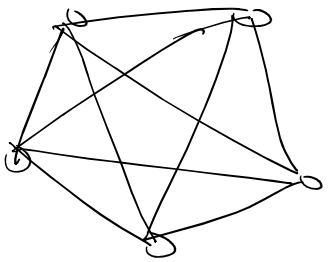
$$= \frac{d(d-1)^k}{d-2} - \frac{d}{d-2} + 1$$

$$< \frac{d(d-1)^k}{d-2}$$

Some commonly seen graphs

(i) Complete graph on n vertices, denoted K_n, and it has all the $\binom{n}{2}$ edges.

K_5



Some definitions

* Independent set: Given a graph G, $S \subseteq V(G)$ is called an independent set if no two vertices in S have an edge between them.

Independence no. of a graph G, denoted $\alpha(G)$ is the size/cardinality of a largest independent set in G.

* Clique: A set $S \subseteq V(G)$ is a clique if every two vertices in S have an edge between them in G.

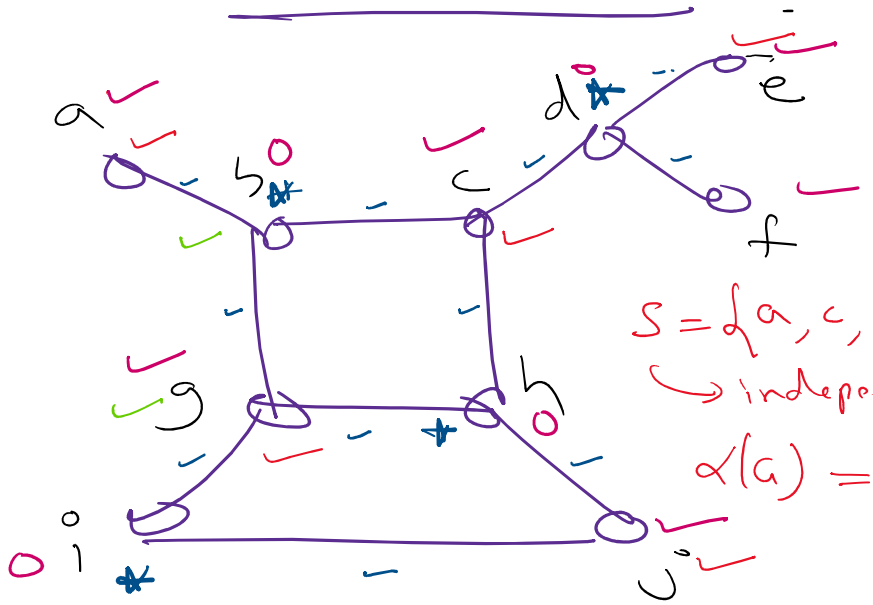
Size/cardinality of a largest clique in G is called clique number of G. It is denoted $\omega(G)$.

* Vertex cover: A set $S \subseteq V(G)$ is a vertex cover of G if every edge in G has an endpoint in S.

A smallest cardinality vertex cover of G is called a minimum vertex cover

* Dominating set : A set $S \subseteq V(G)$ is called a dominating set of G if every vertex in G is either in S or has a neighbor in S.

Example to explain the notion



$S = \{a, c, e, g, j\}$
↳ independent set
$\alpha(G) = 6$

$S' = \{b, g\}$ is a clique.

$S'' = \{b, d, h, i\}$ is a vertex cover.
Is it the smallest one?

$S''' = \{b, d, h, i\}$ is a dominating set.
Is this a smallest dominating set?

↳ Coming back to example 1!

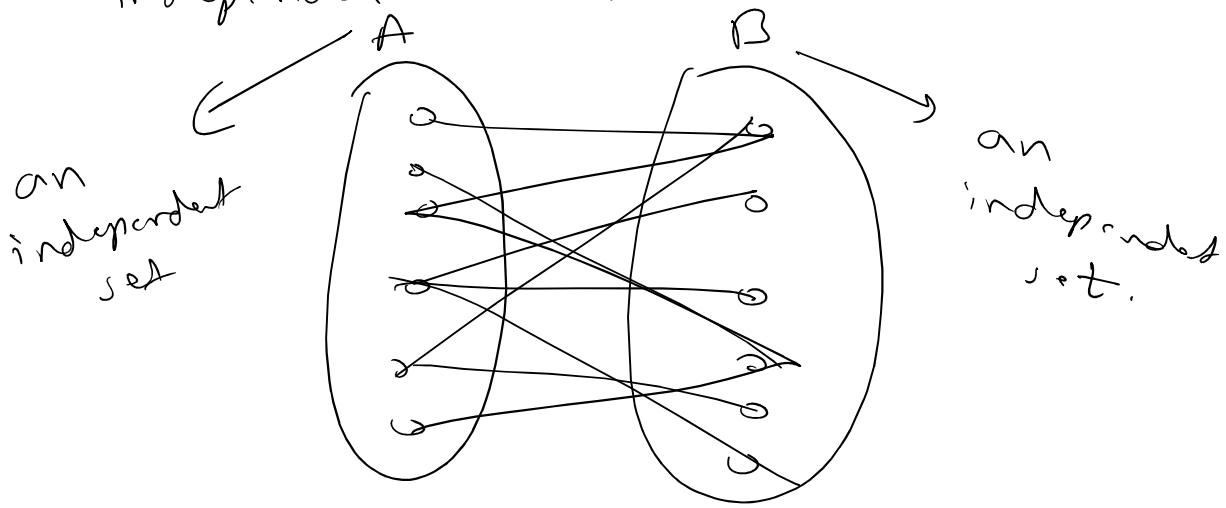
Coming back to examples of common graph

$\alpha(G)$ → max indep. set size
$\beta(G)$ → min vertex cover size
$\gamma(G)$ → largest dominating set size.

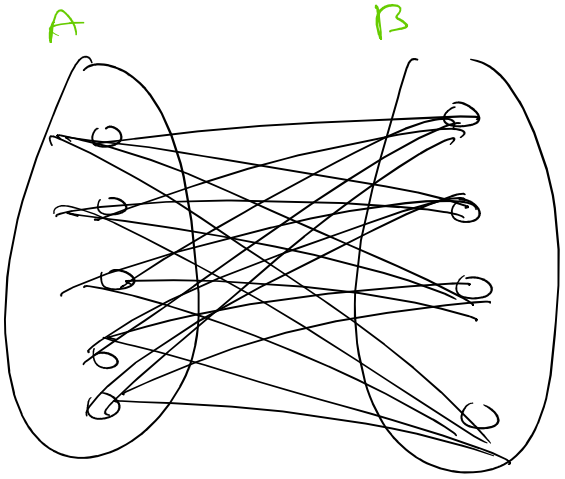
(i) Complete graph on n vertices

$\alpha(G)=1$, $\beta(G)=n-1$, $\gamma(G)=1$

(ii) Bipartite graph: A graph whose vertices can be partitioned into two independent sets.



(iii) Complete bipartite graph



$\alpha(G) = \max(|A|, |B|)$

$\beta(G) = \min(|A|, |B|)$

$\gamma(G) = 2$

(iv) Path graph on n vertices

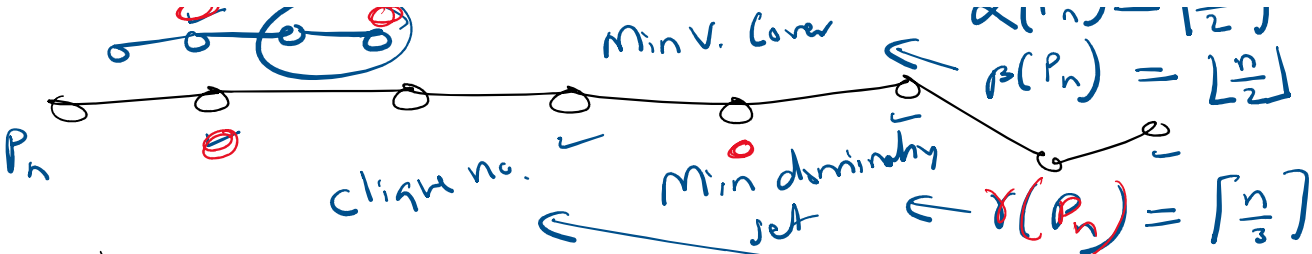


Max indep. set

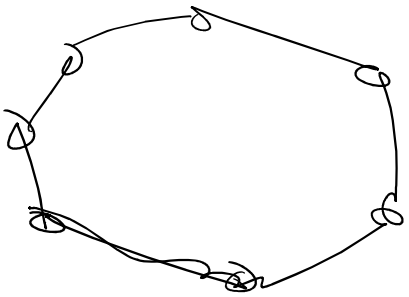
Min V. Cover

$\alpha(P_n) = \lceil \frac{n}{2} \rceil$

$\beta(P_n) = \lfloor \frac{n}{2} \rfloor$



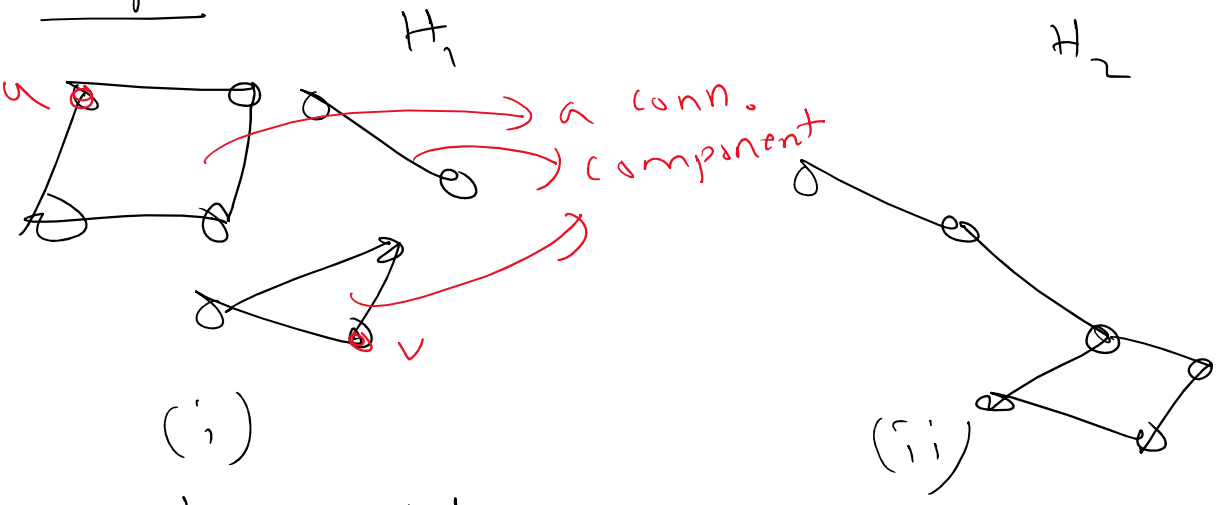
(v) Cycle graph on n vertices $w(P_n) = 2$



Connectivity

A graph G is connected if there is a path between every pair of vertices in G.

Example.



H_1 is not connected.

H_2 is connected.

Defn.! A graph G is k-vertex-connected (edge) if it cannot be disconnected by removing less than k vertices (edges).

Convention: A single vertex graph is

Convention: A single vertex graph is disconnected.

Defn 2: The vertex connectivity of a graph G, denoted $\kappa(G)$, is the maximum k such that G is k-vertex connected.

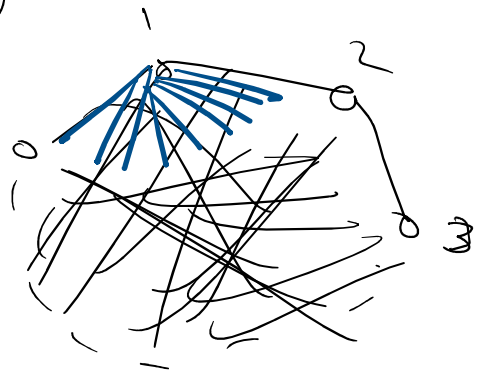
k-vertex connected $\Rightarrow (k-1)$-vertex conn.

minimum no. of vertices to be deleted to disconnect G.

Examples:

Complete graph (K_n)

n



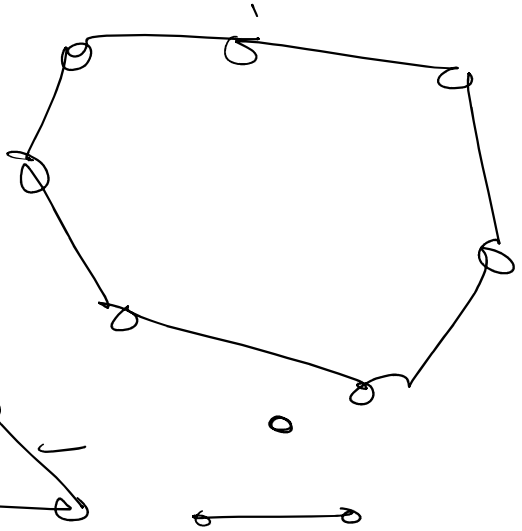
vertex connectivity

$\kappa(K_n) = n-1$

$\lambda(K_n) = n-1$

$\kappa(C_n) = 2$

$\lambda(C_n) = 2$



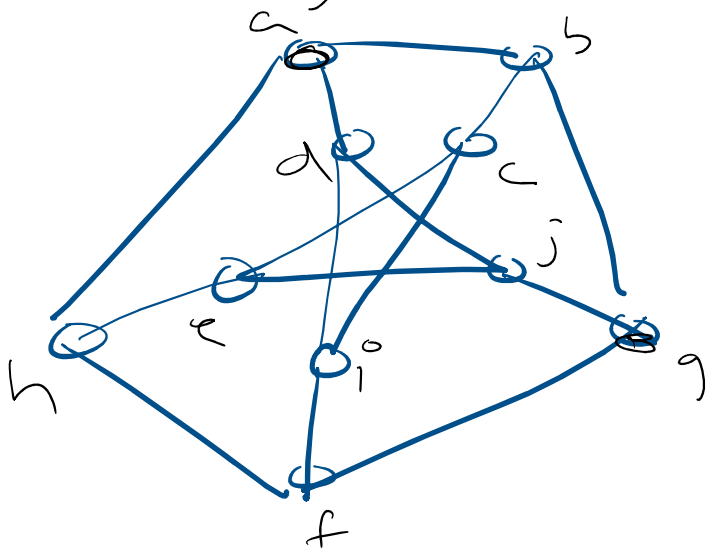
C_n

Prop: Let G be a connected graph on n vertices. Then, there exists an

ordering of its vertices, say $v_1, v_2, \ldots, v_n$, such that $G[\{v_1, \ldots, v_i\}]$ is connected, for all $2 \le i \le n$.

Proof:

Notation: For any $S \subseteq V(G)$, we use $G[S]$ to denote the subgraph of G induced by S.



a, g, f, b, j - - .

$\phi = (\quad) \wedge (\bar{x}_1 \vee \bar{x}_4 \vee x_5) \wedge (x_1 \vee x_2 \vee \bar{x}_3) \wedge (\quad)$. —

n variables.

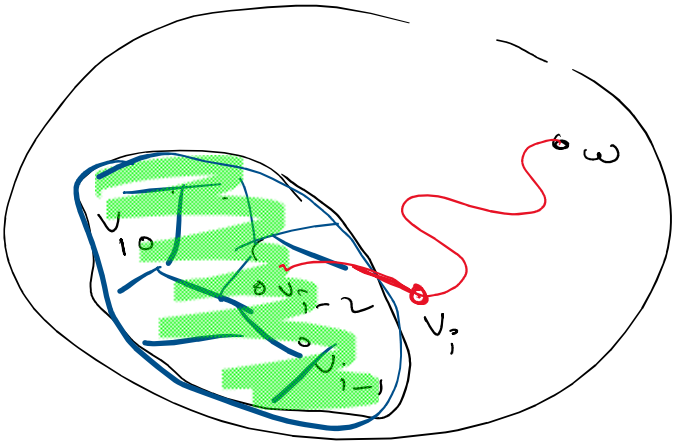
E.T.H. $\Rightarrow$

$2^{O(n)}$ time

2^{cn}, when c is a const.

$1-\delta^n = 2^{0.9n}$.

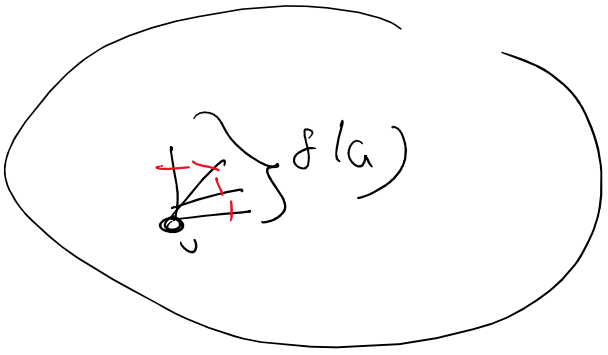
Proof of prop:

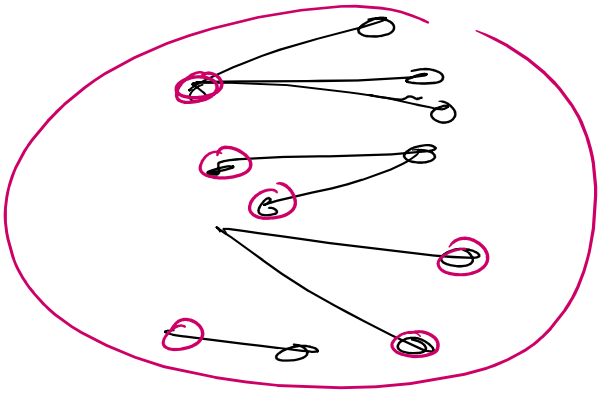


Prop: $\delta(G) \geq \chi(G) \geq \kappa(G)$

min degree ① edge conn. ② vertex conn.

Proof: Proving ①



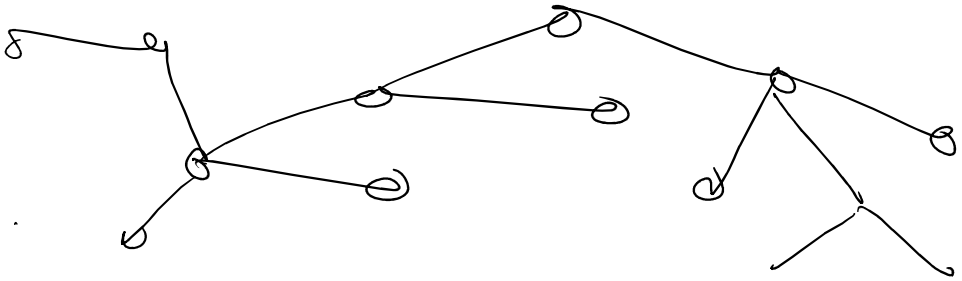


Trees.

An acyclic graph is called a FOREST.

A connected acyclic graph is called a TREE.

example.



Theorem: TFAE (The Following Are Equivalent)

(i) T is a tree,

(ii) Any two vertices of T are linked by a unique path in T;

(iii) T is minimally connected, i.e., T − e is disconned for every edge e in T.

(iv) T is maximally acyclic, i.e. T contains no cycle but T + xy contains a cycle, for every non-adjacent pair x, y.

Proof:

Understanding minimal/maximal vs minimum/maximum

Example

/ maximum.

Example.



$S' = \{b\}$ is a minimum V.C

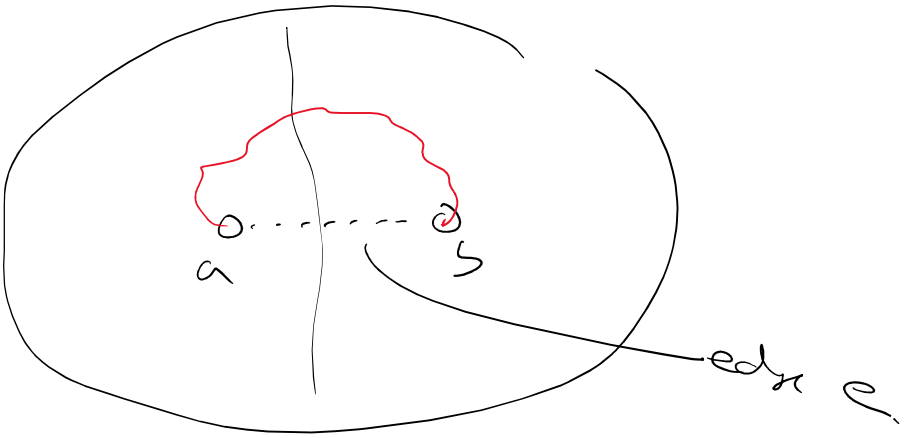
$T' = \{a, c\}$ is a maximum I.S.

$S = \{a, c\}$ is a minimal vertex cover

and

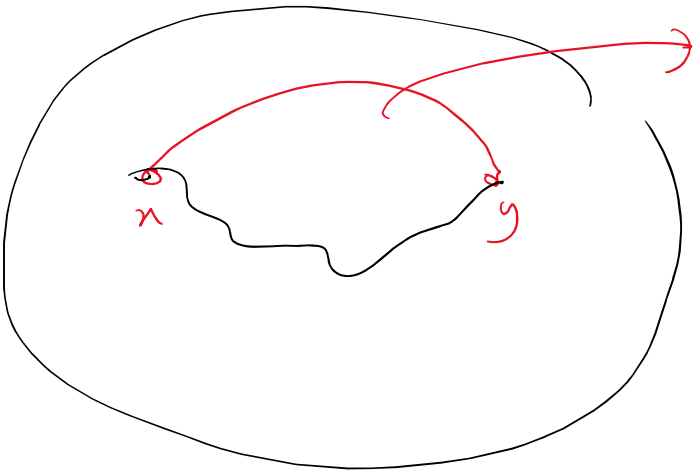
$T = \{b\}$ is a maximal independent set

(ii) $\Rightarrow$ (iii)



(iii) $\Rightarrow$ (iv)

Suppose T is not maximally acyclic.

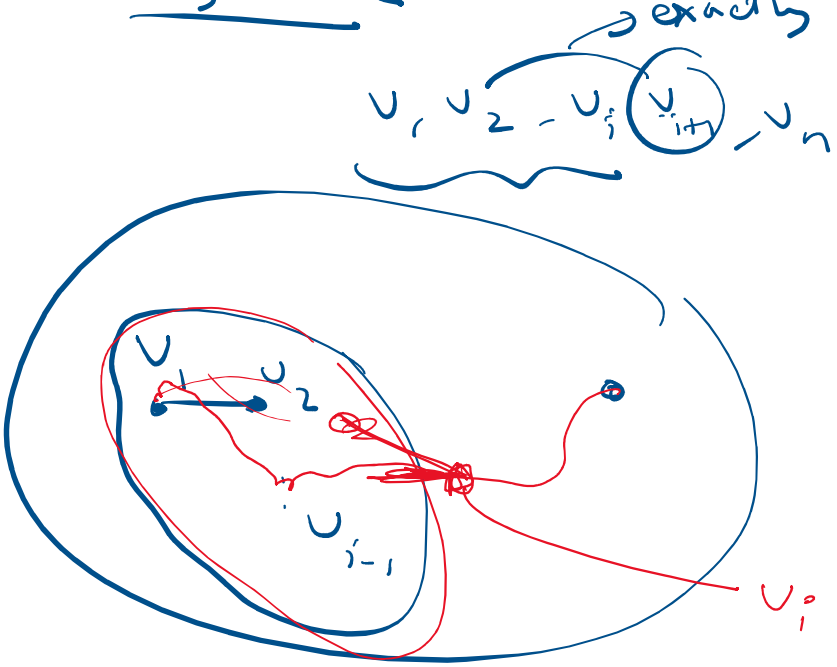


(iv) $\Rightarrow$ (i)

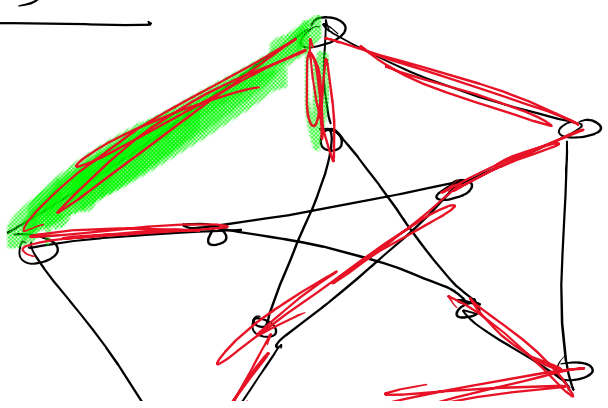
OR

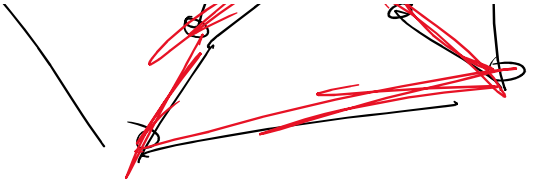
maximally acyclic $\Rightarrow$ connected

Corollary 1.5.2

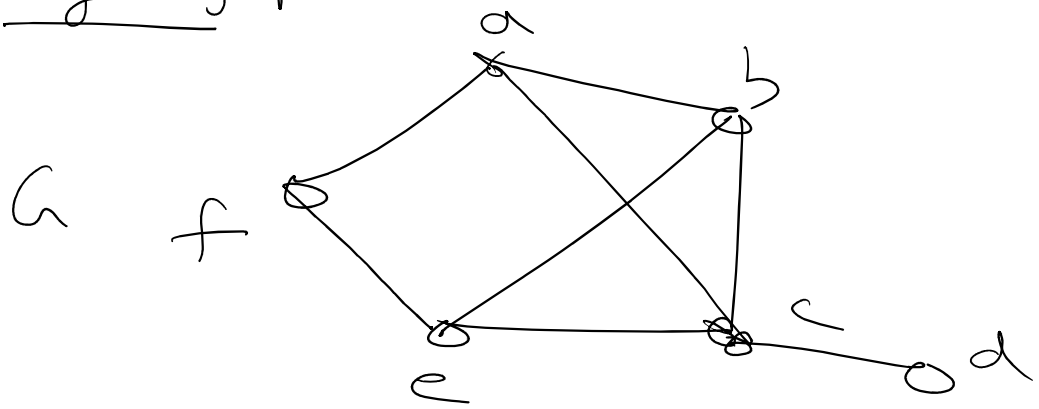


Spanning tree

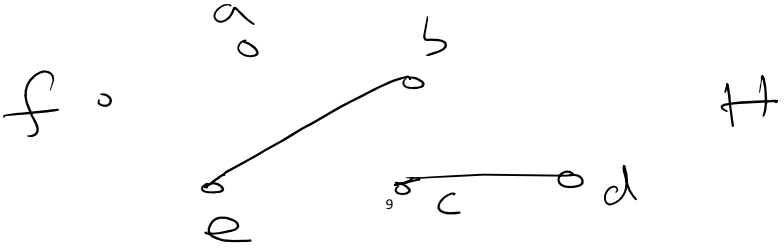




Spanning subgraph



A spanning subgraph of G is a subgraph of G that has all the vertices of G.

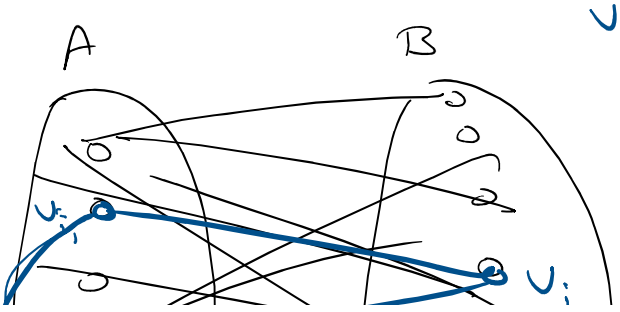


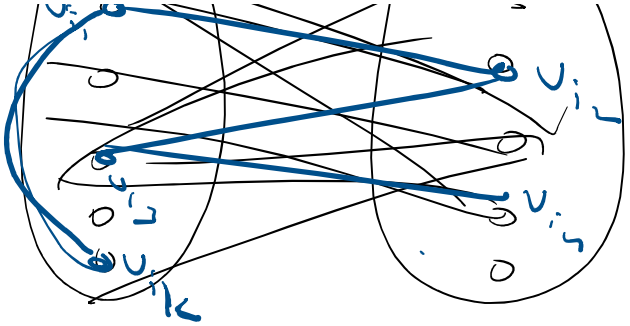
H is a spanning subgraph of G.

* Every connected graph has a spanning subgraph that is a tree.

Prop: A graph is bipartite iff it has no odd cycles.

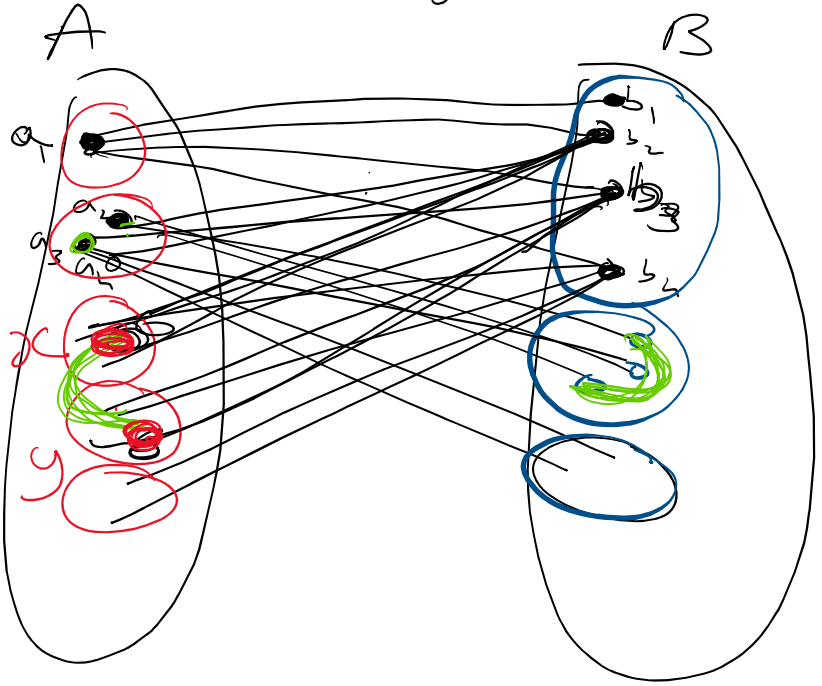
Proof: $\Rightarrow$





A, B are independent sets.

$\Leftarrow$ no odd cycle $\Rightarrow$ bipartite.

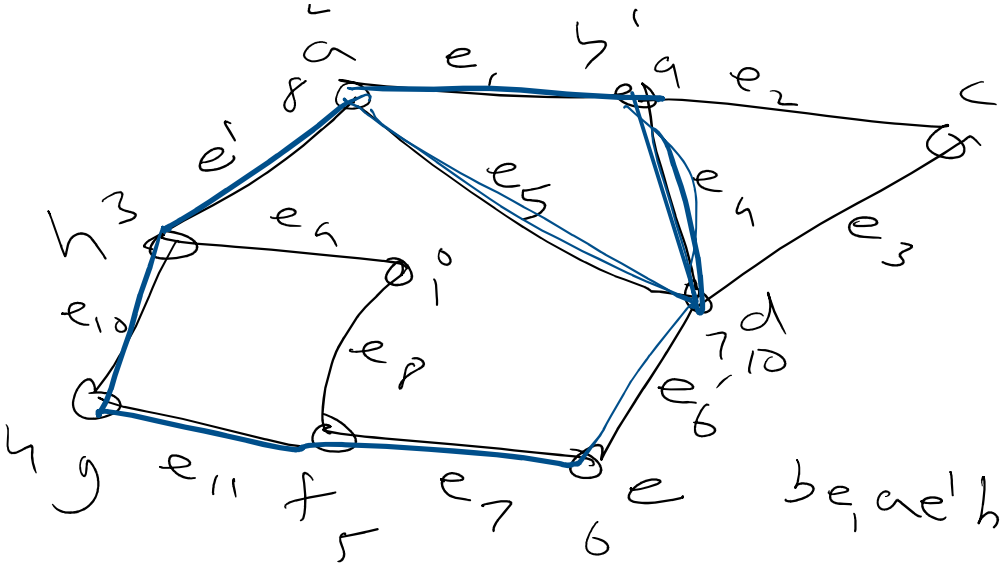


Defn. A walk in a graph is an alternating sequence of vertices and edges say

$v_0 e_0 v_1 e_1 v_2 e_2 \cdots e_{k-1} v_k$, where

$e_i = \{v_i, v_{i+1}\}$, $0 \le i \le k-1$. It is called a closed walk if $v_0 = v_k$.





Eulerian tour

Prop. A connected graph is Eulerian iff every vertex has even degree

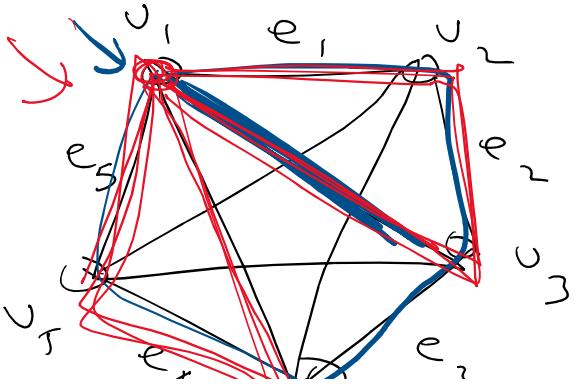
Proof:

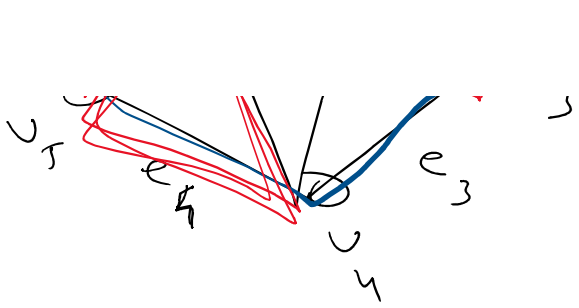
$\Rightarrow$ Eulerian $\Rightarrow$ even degree for all vertices

$\Leftarrow$ with no edge repetitions.

Take the longest walk possible.

$v_0 e_1 v_1 e_2 v_2 e_3 \cdots e_{k-1} v_k$





3Q.

